Anomaly detection in time series :-

Q. what is anomaly detection ?

→ Identify the rare events that that significantly from the majority of the data.

used in wide range of real applications, from manifacturing to healthcare.

Q why do anomaly detection :-

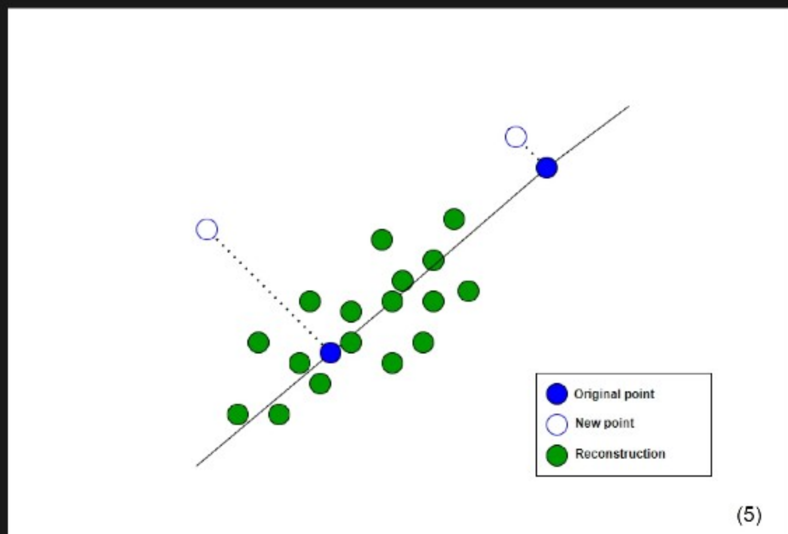
→ unexpected events can be caused by production faults or system defects.

→ outliers can affect the performance of forecasting models.

Types :- ① Point wise anomaly detection

https://www.infoq.com/articles/anomaly-detection-ml-net/
https://www.infoq.com/articles/anomaly-detection-ml-net/

② Patternwise anomaly detection.



Pointwise anomaly detection

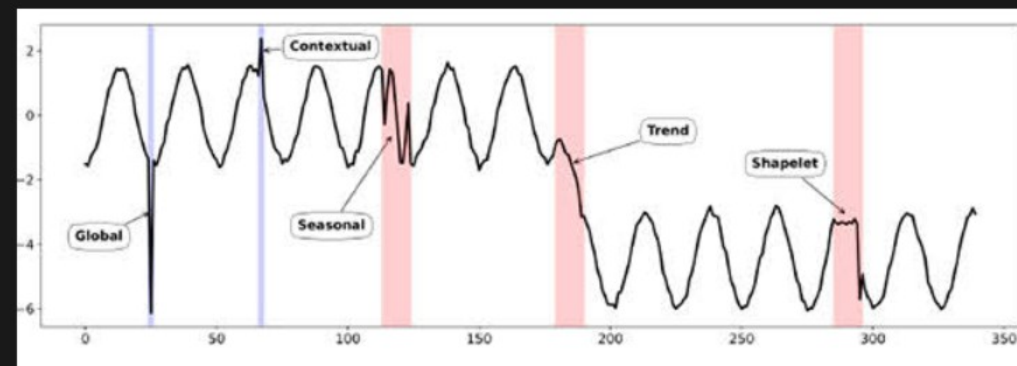


Figure 2: An illustration (borrowed from Darban *et al.* [15]) of the taxonomy of anomaly types suggested by Lai *et al.* [13]

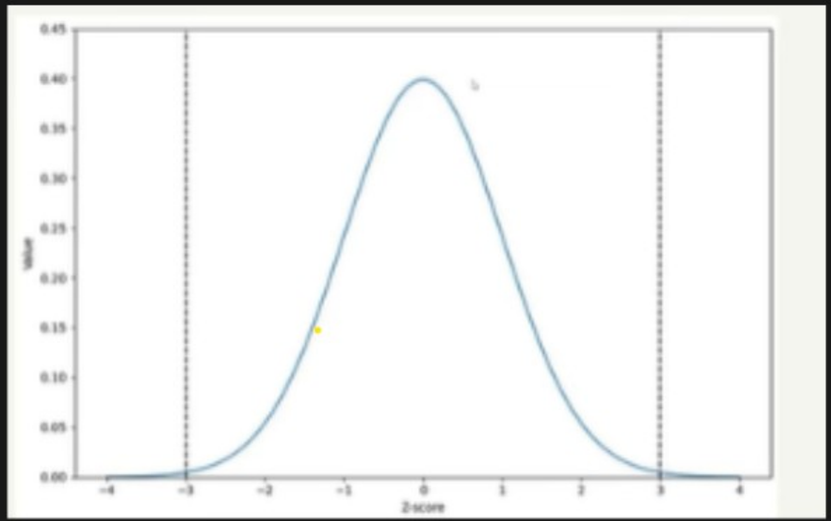
Anlyzing past patterns.

* mean absolute deviation :-

when the data is normally distubuted we can reasonably conclude that points at each tail are outliers.

Z Score :

$$\boxed{Z = \frac{x - \mu}{\sigma}}$$



mean absoulate deviation

* Robust Z score methode

$$MAD = \text{median} \, |x_i - \tilde{x}|$$

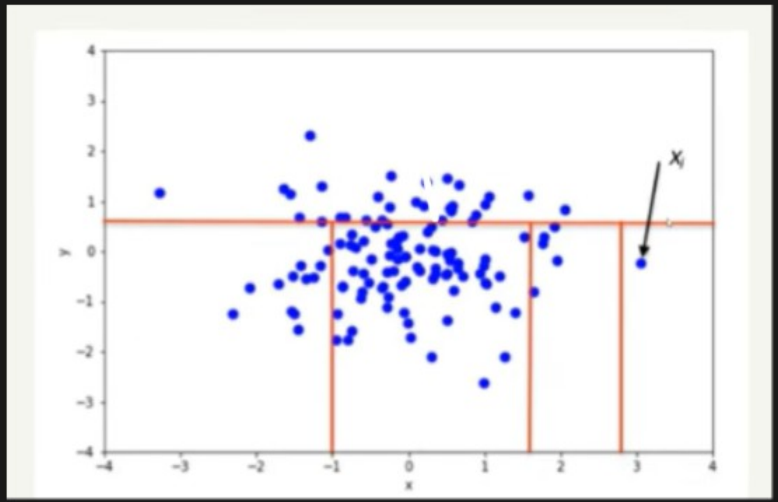
$$\boxed{m_i = \frac{0.675\,(x_i - \tilde{x})}{MAD}}$$

$mad \neq 0$

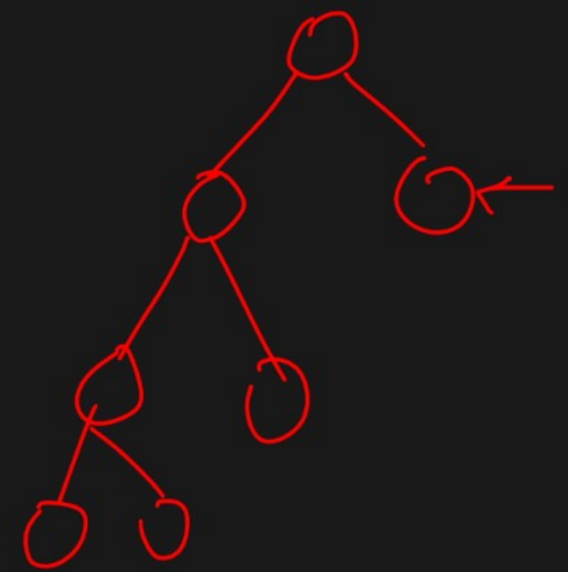
* Robust Z-score works only if data is close to a normal distribution.

- The MAD is not equal to '0' (happens when more than 50% of the data has same value)

Isolation Forest



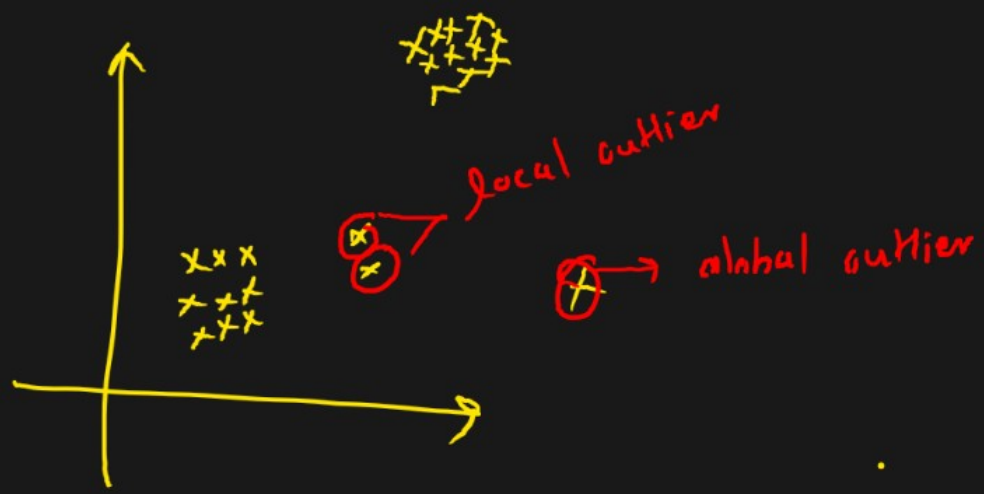
Isolation trees are created



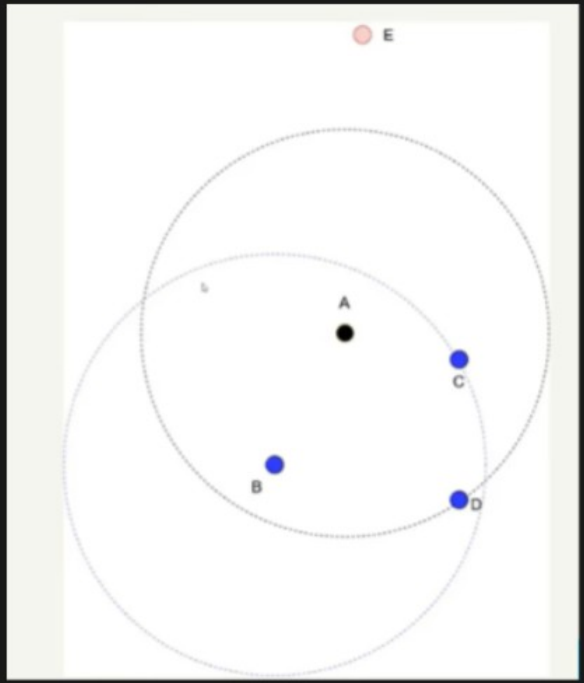
we try to split each data point with leaf nodes.

Sk-learn → we can implement using sk learn

Local Outlier Factor :



→ Reachability distance



local outlier factor.

how we actually conclude

local outlier factor :

→ if lof closer to 1 or smaller than 1 inlier ✓

→ lof greater than 1 outlier ✗

Thank you !!

~ Rohan Adhav